FIRMWARE SECURITY AND VULNERABILITY ANALYSIS IN EMBEDDED IOT SYSTEMS

A SEMINAR REPORT

submitted by

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CERTIFICATE

This is to certify that the seminar report entitled ***"Firmware Security and Vulnerability Analysis in Embedded IoT Systems"*** *submitted by* ***XYZ (AJC21CSXYZ)*** *to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering during the year 2026–2027, is a bonafide work carried out by him/her under my guidance and supervision.*

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ABSTRACT

Integrity is the practice of being honest and showing a consistent and uncompromising adherence to strong moral and ethical principles and values. In ethics, integrity is regarded as the honesty and truthfulness or accuracy of one's actions. Integrity can stand in opposition to hypocrisy, in that judging with the standards of integrity involves regarding internal consistency as a virtue, and suggests that parties holding within themselves apparently conflicting values should account for the discrepancy or alter their beliefs. The word integrity evolved from the Latin adjective integer, meaning whole or complete. In this context, integrity is the inner sense of "wholeness" deriving from qualities such as honesty and consistency of character. As such, one may judge that others "have integrity" to the extent that they act according to the values, beliefs and principles they claim to hold.

An individual's value system provides a framework within which the individual acts in ways which are consistent and expected. Integrity can be seen as the state or condition of having such a framework, and acting congruently within the given framework.One essential aspect of a consistent framework is its avoidance of any unwarranted (arbitrary) exceptions for a particular person or group—especially the person or group that holds the framework. In law, this principle of universal application requires that even those in positions of official power be subject to the same laws as pertain to their fellow citizens.

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ABBREVIATIONS

WAT	Wave Atom Transformation
LPQ	Local Phase Quantization
CNN	Convolutional Neural Network
LBP	Local Binary Pattern
PCB	Printed Circuit Board
BSIF	Binarized Statistical Image Features
LivDet	Liveness Detection
VGG	Visual Geometry Group

CHAPTER 1

INTRODUCTION

1.1 Overview

Biometric security is a mechanism which is used to authenticate and ....

1.2 Motivation and Objectives

..To detect these fingerprint spoofing .... etc.

1.3 Report Organization

This section gives the organization of the report work. Overall, this report work gives a clear and thorough study about the existing study of the proposed system and gives a description about the implementation details. The second chapter gives a detailed description about the literature survey. Third chapter gives a brief idea about the implementation details including hardware and software requirements, outline of the proposed methodology. Fourth chapter depicts the Results and Discussions and fifth Chapter gives the Conclusion and Future Work. The Appendix shows the Screen Shots and Sample Code of the proposed systems.

CHAPTER 2

LITERATURE SURVEY

2.1 Discussion of earlier/older works

The Fingerprint Recognition System has ubiquitous deployment in many day-to-day applications, such as financial transactions, international border security, unlocking a smart phone[1]

Introduction The Fingerprint Recognition System has ubiquitous deployment in many day-to-day applications, such as financial transactions,[4] international border security, unlocking a smart phone, etc.

2.2 Create appropriate subheadings

This paper proposes a method to obtain the discriminant features for fingerprints whether they are real or spoofed, by performing feature level extractions [10]. The fingerprint image is preprocessed using five preprocessing operations. The preprocessing operations were selected which have the lowest validation errors.

1. Image reduction with different ratios Images are reduced using "Bilinear Interpolation". They are used to verify the effect of resolution on the classifier performance.

2. Region Of Interest (ROI) Many fingerprints from some datasets are not centered; background represents large part of image. To input the classification system the largest area that comprises foreground fingerprints, a simple ROI method is used.

They are used to verify the effect of resolution on the classifier performance.They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Add required subtopics as per paper

3.1.1 Xyz

Research Methods: Description of the methods and techniques used in the paper.
Data Collection: How data was gathered, if applicable.
Experimental Setup: Details of any experiments or simulations conducted.
Key Topics: Detailed discussion of the main topics covered in the paper.
Figures/Tables: Include and describe any figures, tables, or charts used in the paper for better understanding.

MATLAB supports graphical user interface (GUI) features for developing applications. For graphically designing these GUIs, MATLAB includes a GUIDE (GUI development environment) which also..

3.2 Xy Z

3.2.1 Xy Z

From these, only Digital-Persona was selected for the training and testing process in order to reduce the time required for processing. It consists of 1000 live images in both training as well as testing sets whereas fake images are obtained from four materials ie. Ecoflex

3.2.2 Xy Z

Wave Atom Transformation for Image Enhancement The fingerprint images obtained from the LivDet 2015 datasets may contain some noise.... and assigned it with two values 1 and 2. If the predicted value is 1, then it is considered as LIVE else it is FAKE.

CHAPTER 4

RESULTS AND DISCUSSIONS

This section gives as a detailed analysis of the proposed approach. Findings: Summary of the key findings or results presented.
Summary of Key Findings: Major takeaways from the seminar.
Analysis: Critical analysis of the information presented, including insights and implications.
We evaluated the performance on three scenarios to compare which one provides better performance.

Evaluation of CNN with feature vector as input In this setting, the fingerprint images in the dataset are firstly enhanced using Wave Atom Transformation method in order to remove the unwanted noise from the image pixels..... It took only 00:00:20 hrs at the time of testing whereas other two methods took more time. Figure 4.1 shows the gradient value obtained in each case. The performance Table 4.1 depicts that Ecoflex 00-50

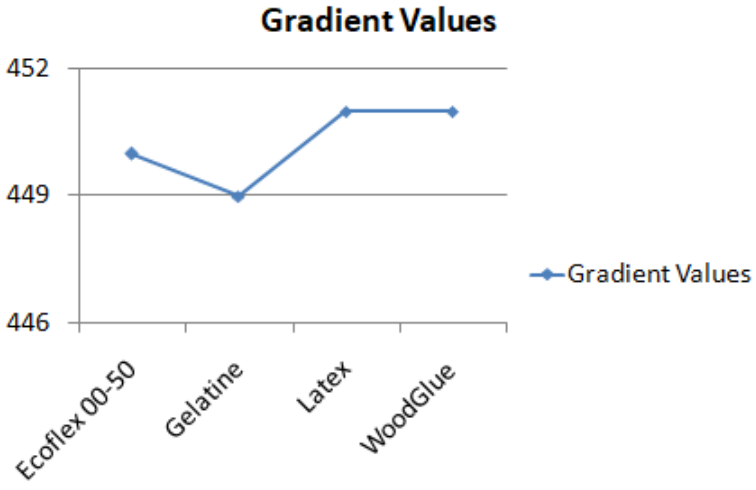


Figure 4.1: Graphical representation of gradient value obtained by each materials in the (ii) scenario

Table 4.1: Correct Rate and Error Rate obtained for CNN with feature vector as input

Materials Used (Live and Fake)	CR	ER
Ecoflex 00-50	0.788	0.211
Gelatine	0.808	0.192
Latex	0.802	0.198
WoodGlue	0.808	0.191

CHAPTER 5

CONCLUSION AND FUTURE WORK

The main limitation of many of the existing anti-spoof methods is the poor quality of the datasets obtained. The spoofing materials of poor image quality decreases the overall performance of the system. So we need either a best hardware for image acquisition or a good image processing method to obtain better results , but most of the hardware are expensive.

Here we proposed a method ..... The proposed WAT enhanced method yields an average accuracy of 85.8%. The WAT enhanced image with minutiae features extracted, achieved an average accuracy of 79.8%. We also computed the performance of CNN without any enhancement on image as input and obtained an average accuracy of 91.6%.

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APPENDIX A

SCREEN SHOTS

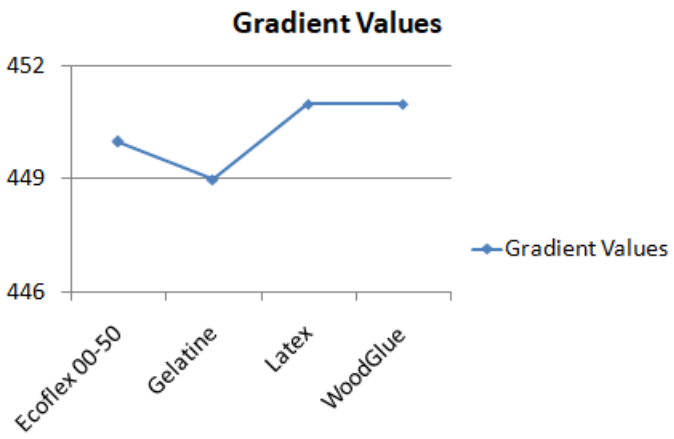


Figure A.1: Sample visualization / Experimental gradient value

APPENDIX B

SAMPLE CODE

Evaluation and Feature Extraction Script

```
close all;
clear;
clc;

% 1. Read input image file
[filename, pathname] = uigetfile('*.png', 'Select an image', 'Dataset\Testing');
I = imread(fullfile(pathname, filename));
addpath('Waveatom');
val = wave(I);

% 2. Extract minutiae features
[rgbImage, ridge_xy, bifurcation_xy] = minutiaeextract(val);

% 3. Model prediction
if cnnPredict == 1
    uiwait(msgbox(sprintf('LIVE'), 'RESULT'));
else
    uiwait(msgbox(sprintf('FAKE'), 'RESULT'));
end
```

Feature Extraction for Dataset Materials

```
clc;
clear;

load fpsFs;
```

```
FList = ReadImageNames('DataSet');
Index = contains(FList, 'Testing\Fake\Ecoflex 00-50');
TestF_name_G = FList(Index, :);
TestF_G = fpsFs(Index, :);

net = trainSoftmaxLayer(TrainF', Ts);
[~, cnn_predict] = max(net(TestF'));
cp = classperf(TestL, cnn_predict);
```